

Intelehealth OpenMRS Platform Installation & Upgrade Guide

Overview

This guide provides a complete step-by-step installation process for:

- OpenMRS Platform
- OpenMRS Reference Application
- Intelehealth Middleware
- Intelehealth Doctor Web App
- Intelehealth Backend Services
- Supporting Python Applications
- OpenMRS Upgrade Procedures

The instructions are written for Ubuntu-based Linux servers.

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1. System Requirements

Recommended server configuration:

Component	Recommended
OS	Ubuntu 20.04+
RAM	8 GB minimum
CPU	4 Cores
Disk	100 GB SSD
Java	OpenJDK 8
Node.js	v24.12.0
MySQL	5.7
Tomcat	8.5.100

2. Install Java

Update system packages:

```
sudo apt-get update
```

Install OpenJDK 8:

```
sudo apt-get install -y openjdk-8-jdk
```

Verify installation:

```
java -version
```

Expected output should include Java 8.

3. Install Docker & MySQL 5.7

Note: MySQL 5.7 is deprecated. However, this setup uses MySQL 5.7 for compatibility with the application stack.

3.1 Install Docker

Install Docker:

```
sudo apt-get install -y docker.io docker-compose
```

Enable and start Docker:

```
sudo systemctl enable docker  
sudo systemctl start docker
```

Verify Docker installation:

```
docker --version  
docker-compose --version
```

3.2 Create MySQL Docker Directory

```
mkdir -p ~/mysql-docker/config  
cd ~/mysql-docker
```

3.3 Configure MySQL

Create the MySQL configuration file:

```
nano config/my.cnf
```

Paste the following configuration:

```
[mysqld]  
# Performance tuning  
innodb_buffer_pool_size = 1G  
innodb_log_file_size = 256M  
innodb_flush_log_at_trx_commit = 1  
innodb_file_per_table = 1  
query_cache_type = 1
```

```
query_cache_size = 64M

max_allowed_packet=64M
bind-address = 0.0.0.0

# Connection tuning
max_connections = 200
wait_timeout = 3600
interactive_timeout = 3600

# Logging and Monitoring
slow_query_log = 1
slow_query_log_file = /var/lib/mysql/mysql-slow.log
long_query_time = 2

# Security
skip-symbolic-links = 1

[client]
max_allowed_packet=64M
```

Save and exit.

3.4 Create Docker Compose File

Create the compose file:

```
nano docker-compose.yml
```

Paste the following:

```
services:
  mysql:
    image: fahadkabir/mysql:5.7
    container_name: mysql
    restart: unless-stopped
    environment:
      MYSQL_ROOT_PASSWORD: ${MYSQL_ROOT_PASSWORD:-openmrs}
      MYSQL_DATABASE: ${MYSQL_DATABASE:-openmrs}
      MYSQL_USER: ${MYSQL_USER:-openmrs}
      MYSQL_PASSWORD: ${MYSQL_PASSWORD:-openmrs}
    ports:
      - "3306:3306"
```

```
volumes:
  - db_data:/var/lib/mysql
  - ./config:/etc/mysql/conf.d
logging:
  driver: "json-file"
  options:
    max-size: "10m"
    max-file: "3"

volumes:
  db_data:
    driver: local
```

Save and exit.

3.5 Start MySQL Container

```
docker-compose up -d
```

Verify container is running:

```
docker ps
```

4. Restore Existing Database

If you already have an OpenMRS database backup:

Copy the SQL backup file to the server.

Restore database:

```
mysql -u root -p openmrs < backup.sql
```

If MySQL is running inside Docker:

```
docker exec -i mysql mysql -u root -popenmrs openmrs < backup.sql
```

5. Install Apache Tomcat 8.5.100

Move to a working directory:

```
cd ~
```

Download Tomcat:

```
wget https://archive.apache.org/dist/tomcat/tomcat-8/v8.5.100/bin/apache-tomcat-8.5.100.zip
```

Install unzip utility:

```
sudo apt-get install -y unzip
```

Extract Tomcat:

```
unzip apache-tomcat-8.5.100.zip
```

Move Tomcat to `/opt`:

```
sudo mv apache-tomcat-8.5.100 /opt/tomcat8
```

Grant permissions:

```
sudo chmod +x /opt/tomcat8/bin/*.sh
```

6. Install OpenMRS 2.1.4

Download OpenMRS WAR:

```
wget https://sourceforge.net/projects/openmrs/files/releases/OpenMRS_Platform_2.1.4/openmrs.war/download
```

Rename and move WAR file:

```
mv download openmrs.war  
sudo mv openmrs.war /opt/tomcat8/webapps/openmrs.war
```

7. Install Reference Application 2.8.0

Move to your home directory:

```
cd ~
```

Download Reference Application:

```
wget https://sourceforge.net/projects/openmrs/files/releases/  
OpenMRS_Reference_Application_2.8.0/referenceapplication-addons-2.8.0.zip/  
download
```

Rename ZIP file:

```
mv download refapp-2.8.0.zip
```

Extract files:

```
unzip refapp-2.8.0.zip
```

Create OpenMRS modules directory:

```
mkdir -p ~/.OpenMRS/modules
```

Copy modules:

```
cp -r refapp-2.8.0/* ~/.OpenMRS/modules/
```

8. Install Webservice REST Module

Download module:

```
wget https://addons.openmrs.org/api/v1/addon/org.openmrs.module.webservices-rest/2.31.0/download
```

Remove existing REST module:

```
rm ~/.OpenMRS/modules/webservices.rest-*.omod
```

Rename downloaded file:

```
mv download webservices.rest-2.31.0.omod
```

Move module:

```
mv webservices.rest-2.31.0.omod ~/.OpenMRS/modules/
```

9. Configure and Start Tomcat

Start Tomcat:

```
cd /opt/tomcat8/bin  
./startup.sh
```

Verify Tomcat is running:

```
ps aux | grep tomcat
```

Open the following URL in your browser:

```
http://SERVER_IP:8080/openmrs
```

10. Install Maven 3.6.3

Move to home directory:


```
cd ~
```

Download Maven:

```
wget https://archive.apache.org/dist/maven/maven-3/3.6.3/binaries/apache-maven-3.6.3-bin.zip
```

Extract Maven:

```
unzip apache-maven-3.6.3-bin.zip
```

Move Maven:

```
sudo mv apache-maven-3.6.3 /opt/maven
```

Edit `.bashrc`:

```
nano ~/.bashrc
```

Add the following lines:

```
export M2_HOME=/opt/maven
export PATH=$PATH:$M2_HOME/bin
```

Reload `.bashrc`:

```
source ~/.bashrc
```

Verify Maven:

```
mvn -version
```

11. Setup Intelhealth Middleware

11.1 Clone Repository

```
git clone git@github.com:Intelehealth/intelehealth-middleware.git
```

Move into project:

```
cd intelehealth-middleware/EMR-Middleware
```

11.2 Configure Database Properties

Edit configuration:

```
nano src/main/resources/db_properties.xml
```

Paste the following and update values according to your environment:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE configuration
    PUBLIC "-//mybatis.org//DTD Config 3.0//EN"
    "http://mybatis.org/dtd/mybatis-3-config.dtd">
<configuration>

    <environments default="development">

        <environment id="intelehealthAwsTest">
            <transactionManager type="JDBC"/>
            <dataSource type="POOLED">
                <property name="poolMaximumActiveConnections" value="6"/>
                <property name="poolMaximumIdleConnections" value="3"/>
                <property name="poolMaximumCheckoutTime" value="24000"/>
                <property name="poolPingConnectionsNotUsedFor" value="24000"/>
                <property name="poolPingEnabled" value="true"/>
                <property name="poolPingQuery" value="select 1"/>
                <property name="driver" value="com.mysql.cj.jdbc.Driver"/>
                <property name="url" value="jdbc:mysql://192.168.19.122:3306/
openmrs?autoReconnect=true&useSSL=FALSE"/>

                <property name="username" value="root"/>
            </dataSource>
        </environment>
    </environments>
</configuration>
```

```
        <property name="password" value="openmrs"/>

    </dataSource>
</environment>

</environments>

<mappers>
    <mapper class="com.emrmiddleware.dmo.PatientDMO"/>
    <mapper class="com.emrmiddleware.dmo.VisitDMO"/>
    <mapper class="com.emrmiddleware.dmo.EncounterDMO"/>
    <mapper class="com.emrmiddleware.dmo.ObsDMO"/>
    <mapper class="com.emrmiddleware.dmo.LocationDMO"/>
    <mapper class="com.emrmiddleware.dmo.ProviderDMO"/>
    <mapper class="com.emrmiddleware.dmo.PersonDMO"/>
    <mapper class="com.emrmiddleware.dmo.UserCredentialsDMO"/>
    <mapper class="com.emrmiddleware.dmo.MindmapDMO"/>
</mappers>

</configuration>
```

11.3 Build Middleware

```
mvn clean package -Dmaven.test.skip=true
```

11.4 Configure Tomcat Environment Variables

Create or edit `setenv.sh`:

```
nano /opt/tomcat8/bin/setenv.sh
```

Paste the following and update values according to your environment:

```
export DBHOST='192.168.19.122'
export SUMMARY_LINK='63bf4014-2bbf-4c92-a72f-0309ed18a10e'
export DBURL='jdbc:mysql://192.168.19.122:3306/openmrs?stringtype=unspecified'
export OPENMRS_JDBC_URL='jdbc:mysql://192.168.19.122:3306/openmrs?
stringtype=unspecified'
export DBUSER='root'
export OPENMRS_DB_USER='root'
```

```
export DBPASS='openmrs'  
export OPENMRS_DB_USER_PASSWORD='openmrs'  
export MMBaseURL='https://intelehealth-prod-master-new.mpower-social.com:3004'  
export MMHOST='https://intelehealth-prod-master-new.mpower-social.com:3004'  
export CONFIGURL='https://intelehealth-prod-master-new.mpower-social.com:4004/  
api/config/'
```

Save file.

Grant execute permission:

```
chmod +x /opt/tomcat8/bin/setenv.sh
```

11.5 Deploy Middleware

Copy WAR file:

```
cp target/EMR-Middleware.war /opt/tomcat8/webapps/
```

Restart Tomcat:

```
cd /opt/tomcat8/bin  
./shutdown.sh  
./startup.sh
```

12. Setup Doctor Web Application

12.1 Install NVM

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.3/install.sh | bash
```

Close and reopen terminal.

Verify NVM:

```
nvm --version
```

12.2 Install Node.js

```
nvm install 24.12.0
nvm use 24.12.0
nvm alias default 24.12.0
```

Verify Node.js:

```
node -v
npm -v
```

12.3 Clone Doctor Web Application

```
git clone git@github.com:Intelehealth/intelehealth-doctor-webapp.git
cd intelehealth-doctor-webapp
```

12.4 Create Environment File

Create `.env` file:

```
nano .env
```

Paste the following:

```
PRODUCTION=false
BASE=https://intelehealth-prod-master-new.mpower-social.com
BASE_URL=https://intelehealth-prod-master-new.mpower-social.com/openmrs/ws/rest/
v1
BASE_URL_CORD_APP=https://intelehealth-prod-master-new.mpower-social.com/
openmrs/coreapps/diagnoses
BASE_URL_LEGACY=https://intelehealth-prod-master-new.mpower-social.com/openmrs/
MIND_MAP_URL=https://intelehealth-prod-master-new.mpower-social.com:3004/api
CONFIG_URL=https://intelehealth-prod-master-new.mpower-social.com:4004/api
```

```
NOTIFICATION_URL=https://intelehealth-prod-master-new.mpower-social.com:3004/
notification
SOCKET_URL=https://intelehealth-prod-master-new.mpower-social.com:3004
CAPTCHA_SITE_KEY=YOUR_CAPTCHA_KEY
WEB_RTC_SDK_SERVER_URL=wss://intelehealth-prod-master-new.mpower-social.com:9090
WEB_RTC_TOKEN_SERVER_URL=https://intelehealth-prod-master-new.mpower-social.com:
3030
SITE_KEY=YOUR_SITE_KEY
VAPID_PUBLIC_KEY=YOUR_VAPID_PUBLIC_KEY
AUTH_GATE_WAY_URL=https://intelehealth-prod-master-new.mpower-social.com:3030/
v2/
FIREBASE_API_KEY=YOUR_FIREBASE_API_KEY
FIREBASE_AUTH_DOMAIN=intelhealth-mpower.firebaseio.com
FIREBASE_PROJECT_ID=intelhealth-mpower
FIREBASE_STORAGE_BUCKET=intelhealth-mpower.appspot.com
FIREBASE_MESSAGING_SENDER_ID=971382487208
FIREBASE_APP_ID=YOUR_FIREBASE_APP_ID
SHOW_CAPTCHA=false
FEATURES_LIST=["followUpType", "tnmStaging", "referralFacility",
"priorityOfReferral","visitFollowUp","visitNotes"]
```

12.5 Install Dependencies

```
npm install
```

12.6 Build Application

```
npm run build
```

12.7 Deploy Application to Tomcat

```
cd /opt/tomcat8/webapps
cp -r ~/intelehealth-doctor-webapp/dist/intelehealth-ui ./ROOT
```

Restart Tomcat:

```
cd /opt/tomcat8/bin  
./shutdown.sh  
./startup.sh
```

13. Setup Intelhealth Backend Services

13.1 Install PM2

```
npm install -g pm2@latest
```

Verify PM2:

```
pm2 -v
```

13.2 Clone Backend Repository

```
git clone git@github.com:Intelehealth/intelehealth-backend-service.git  
cd intelehealth-backend-service
```

13.3 Setup Auth Gateway

Move to auth gateway:

```
cd auth-gateway
```

Create environment file:

```
nano .env
```

Configure all required environment variables.

Install dependencies:

```
npm install
```

Start service:

```
pm2 start bin/www --name auth-gateway  
pm2 save
```

Verify:

```
pm2 status
```

13.4 Setup Portal Service

Move to portal:

```
cd ../portal
```

Create environment file:

```
nano .env
```

Configure all required environment variables.

Install dependencies:

```
npm install
```

Start service:

```
pm2 start bin/www --name portal  
pm2 save
```

13.5 Setup WebRTC Service

Move to WebRTC service:

```
cd ../web-rtc
```

Create environment file:

```
nano .env
```

Configure all required environment variables.

Install dependencies:

```
npm install
```

Start service:

```
pm2 start src/index.ts --name=webrtc-ms  
pm2 save
```

13.6 Setup Configuration Microservice

Move to configuration microservice:

```
cd ../configuration-microservice
```

Create environment file:

```
nano .env
```

Configure all required environment variables.

Install dependencies:

```
npm install
```

Start service:

```
pm2 start npm --name configuration-microservice -- run start  
pm2 save
```

14. Setup Python Application

14.1 Install Python Dependencies

```
sudo apt-get install -y python3  
sudo apt-get install -y python3-dev  
sudo apt-get install -y python3-venv
```

Create virtual environment:

```
python3 -m venv venv
```

Activate environment:

```
source venv/bin/activate
```

Install dependencies:

```
pip install -r requirements.txt
```

14.2 Configure Database

Edit `index.py`:

```
nano index.py
```

Update database connection details:

```
mysql.connector.connect(  
    host='localhost',  
    database='openmrs',  
    user='root',  
    password='openmrs'  
)
```

14.3 Start Python Application

```
python index.py
```

15. OpenMRS Upgrade Procedures

This section documents the exact upgrade path.

Upgrade: OpenMRS 2.1.4 → 2.2

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 2: Download OpenMRS 2.2 WAR

Download the OpenMRS 2.2 WAR file.

Step 3: Backup Existing Files

Backup:

- Database
- Existing `openmrs.war`

Step 4: Replace WAR File

```
rm /opt/tomcat8/webapps/openmrs.war  
cp [download_path]/openmrs.war /opt/tomcat8/webapps/
```

Step 5: Update Database URL

Edit runtime properties:

```
nano ~/.OpenMRS/openmrs-runtime.properties
```

Update:

```
connection.url=jdbc:mysql://192.168.19.122:3306/openmrs?  
useSSL=false&autoReconnect=true&sessionVariables=default_storage_engine=InnoDB&useUnicode=true&ch
```

Step 6: Start Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./startup.sh
```

Upgrade Reference Application to 2.9.0

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 2: Download Reference Application 2.9.0

Download and extract the package.

Step 3: Update Modules

```
cd ~/.OpenMRS/modules
```

Delete all modules except:

```
webservices.rest-2.31.0.omod
```

Copy new modules:

```
cp [download_path]/referenceapplication-package-2.9.0/modules/* ./
```

Step 4: Start Tomcat

```
cd /opt/tomcat8/bin/  
./startup.sh
```

Upgrade OpenMRS 2.2 → 2.3

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 2: Download OpenMRS 2.3 WAR

Download OpenMRS 2.3 WAR file.

Step 3: Backup Existing Files

Backup:

- Database

- Existing WAR file
-

Step 4: Replace WAR File

```
rm /opt/tomcat8/webapps/openmrs.war  
cp [download_path]/openmrs.war /opt/tomcat8/webapps/
```

Step 5: Start Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./startup.sh
```

The database will be updated automatically.

Tomcat may fail because of outdated modules.

Step 6: Stop Tomcat Again

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 7: Upgrade Reference Application Modules to 2.10

```
cd ~/.OpenMRS/modules
```

Delete all modules except:

```
webservices.rest-2.31.0.omod
```

Copy modules:

```
cp [download_path]/referenceapplication-package-2.10/modules/* ./
```

Step 8: Start Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./startup.sh
```

Upgrade OpenMRS 2.3 → 2.3.1

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 2: Replace WAR File

```
rm /opt/tomcat8/webapps/openmrs.war  
cp [download_path]/openmrs.war /opt/tomcat8/webapps/
```

Step 3: Start Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./startup.sh
```

Upgrade OpenMRS 2.3.1 → 2.3.2

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 2: Replace WAR File

```
rm /opt/tomcat8/webapps/openmrs.war  
cp [download_path]/openmrs.war /opt/tomcat8/webapps/
```

Step 3: Start Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./startup.sh
```

Upgrade Reference Application to 2.11.0

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 2: Update Modules

```
cd ~/.OpenMRS/modules
```

Delete all modules except:

```
webservices.rest-2.31.0.omod
```

Copy new modules:

```
cp [download_path]/referenceapplication-package-2.11.0/modules/* ./
```

Step 3: Start Tomcat

```
cd /opt/tomcat8/bin/  
./startup.sh
```

Upgrade OpenMRS 2.3.2 → 2.3.3

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 2: Replace WAR File

```
rm /opt/tomcat8/webapps/openmrs.war  
cp [download_path]/openmrs.war /opt/tomcat8/webapps/
```

Step 3: Start Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./startup.sh
```

Tomcat may fail because of the `fhir2` module.

Step 4: Remove FHIR2 Module

```
cd /opt/tomcat8/bin  
sudo ./shutdown.sh
```

Remove module:

```
cd ~/.OpenMRS/modules
rm fhir2-*.omod
```

Restart Tomcat:

```
cd /opt/tomcat8/bin/
sudo ./startup.sh
```

Upgrade OpenMRS 2.3.3 → 2.4

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/
sudo ./shutdown.sh
```

Step 2: Replace WAR File

```
rm /opt/tomcat8/webapps/openmrs.war
cp [download_path]/openmrs.war /opt/tomcat8/webapps/
```

Step 3: Start Tomcat

```
cd /opt/tomcat8/bin/
sudo ./startup.sh
```

Upgrade Reference Application to 2.12.0

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/
sudo ./shutdown.sh
```

Step 2: Update Modules

```
cd ~/.OpenMRS/modules
```

Delete all modules except:

```
webservices.rest-2.31.0.omod
```

Copy modules:

```
cp [download_path]/referenceapplication-package-2.12.0/modules/* ./
```

Step 3: Start Tomcat

```
cd /opt/tomcat8/bin/  
./startup.sh
```

You may encounter Liquibase checksum errors.

Step 4: Fix Liquibase Checksums

Stop Tomcat:

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Run SQL updates:

```
UPDATE openmrs.liquibasechangelog  
SET MD5SUM = '8:a338427d100eb59b39015c6773b85dba'  
WHERE MD5SUM = '8:3a65e41b9238d0feab36ef2e7c0022d3';  
  
UPDATE openmrs.liquibasechangelog  
SET MD5SUM = '8:f2e2cb0a28fe3b6a94a95e49661a5bb7'  
WHERE MD5SUM = '8:f2e2cb0a28fe3b6a94a95e49661a5bb7';
```

Step 5: Start Tomcat

```
cd /opt/tomcat8/bin/  
./startup.sh
```

Upgrade OpenMRS 2.4 → 2.4.2

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 2: Replace WAR File

```
rm /opt/tomcat8/webapps/openmrs.war  
cp [download_path]/openmrs.war /opt/tomcat8/webapps/
```

Step 3: Start Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./startup.sh
```

Upgrade OpenMRS 2.4.2 → 2.4.3

Step 1: Stop Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh
```

Step 2: Replace WAR File

```
rm /opt/tomcat8/webapps/openmrs.war  
cp [download_path]/openmrs.war /opt/tomcat8/webapps/
```

Step 3: Start Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./startup.sh
```

Step 4: Upgrade FHIR2 Module

Login to OpenMRS as administrator.

Navigate to:

```
Administration → Manage Modules
```

Update:

```
fhir2-2.0.0.omod
```

Step 5: Restart Tomcat

```
cd /opt/tomcat8/bin/  
sudo ./shutdown.sh  
sudo ./startup.sh
```

16. Troubleshooting

Check Tomcat Logs

```
tail -f /opt/tomcat8/logs/catalina.out
```

Check PM2 Logs

```
pm2 logs
```

Check Running Services

```
pm2 status
```

Check Open Ports

```
sudo netstat -tulpn
```

Restart Tomcat

```
cd /opt/tomcat8/bin  
./shutdown.sh  
./startup.sh
```

Restart PM2 Applications

```
pm2 restart all
```

Final Verification

After successful installation:

Service	Port
OpenMRS	8080
Portal	3004
Auth Gateway	3030
Configuration Microservice	4004
WebRTC	3000
MySQL	3306

Verify all services are running:

```
pm2 status  
docker ps
```

Open OpenMRS in browser:

```
http://SERVER_IP:8080/openmrs
```

Installation is complete.